Lecture 38

Hamcycle $\leq_p$ *TSP*, Approximation Algorithms, *OptVertexCover*

$Hamcycle \leq_p TSP$

Hamcycle $\leq_p$ *TSP*

- *Hamcycle* $= \{\langle G \rangle \mid G$ is an undirected graph with a hamiltonian cycle$\}$

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Hamiltonian cycle

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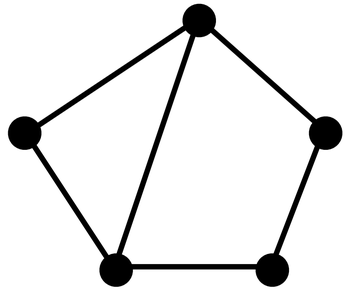
G



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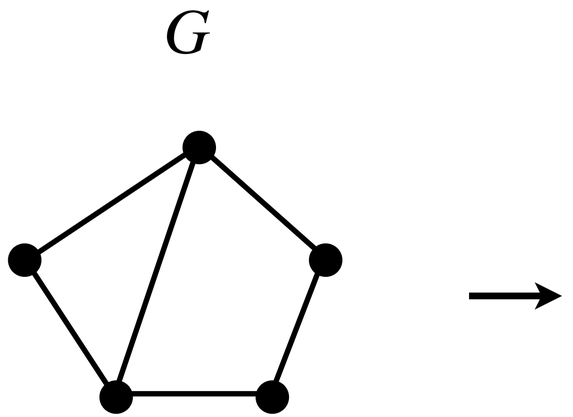

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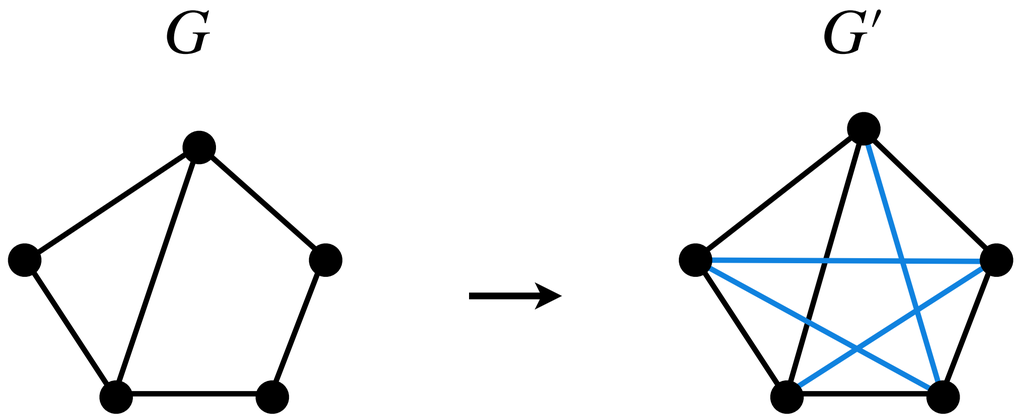

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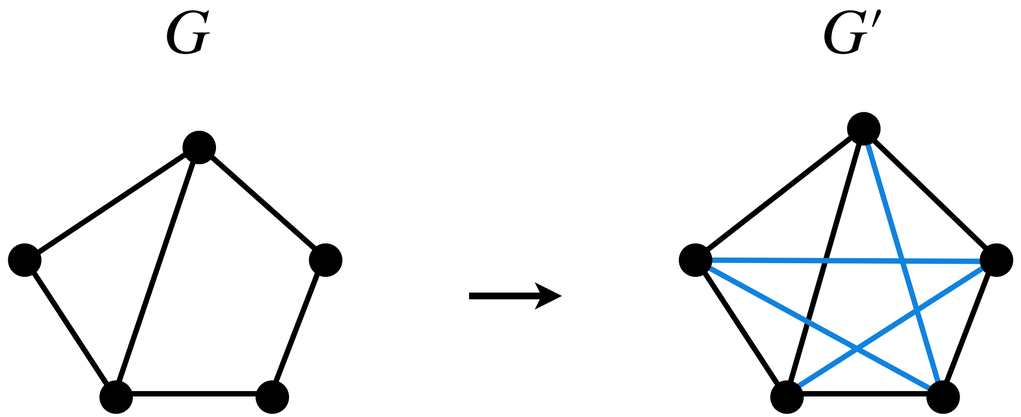

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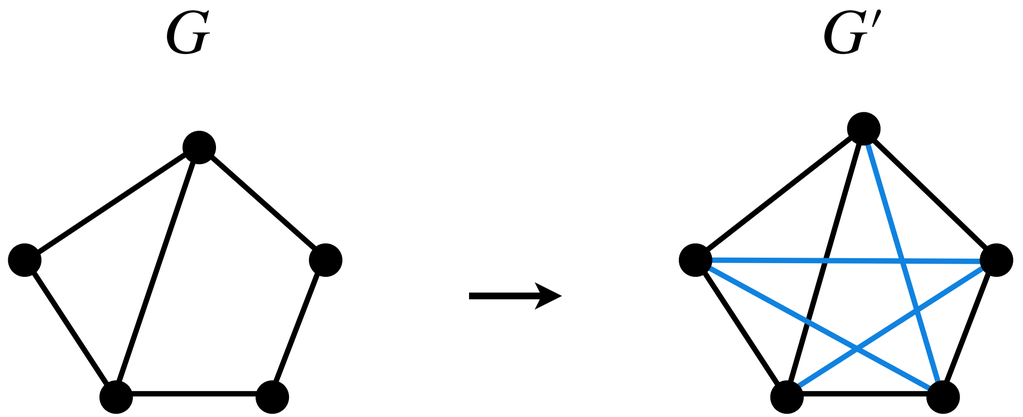

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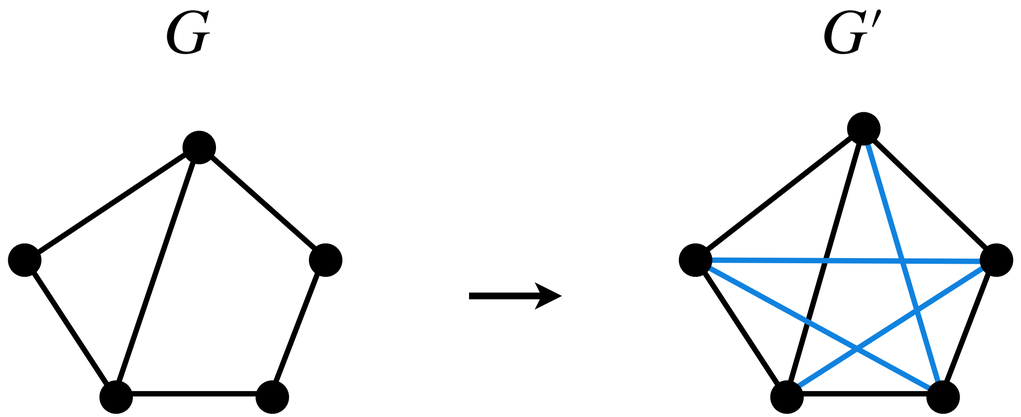

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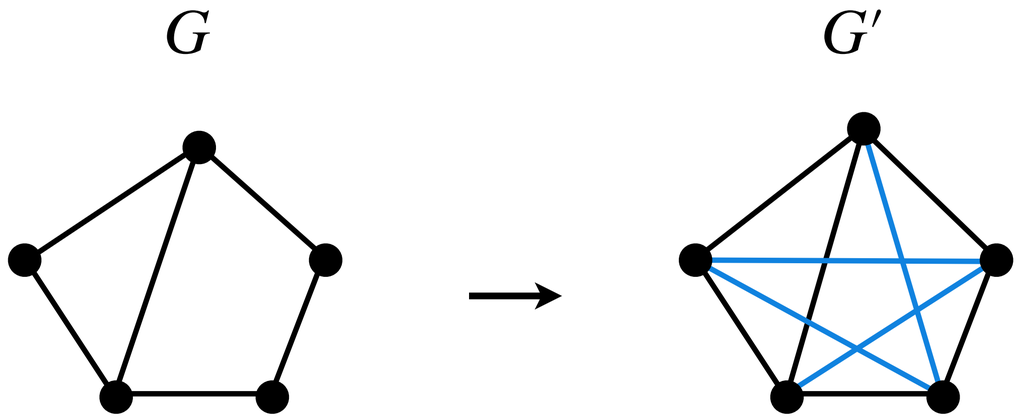

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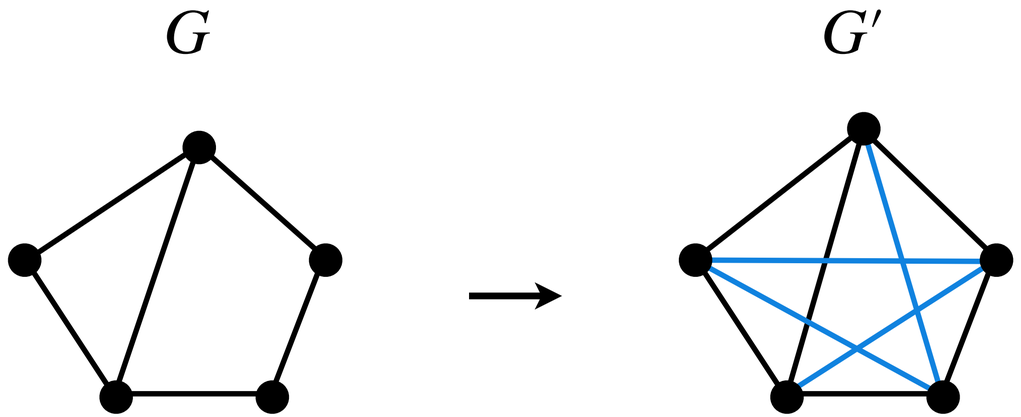

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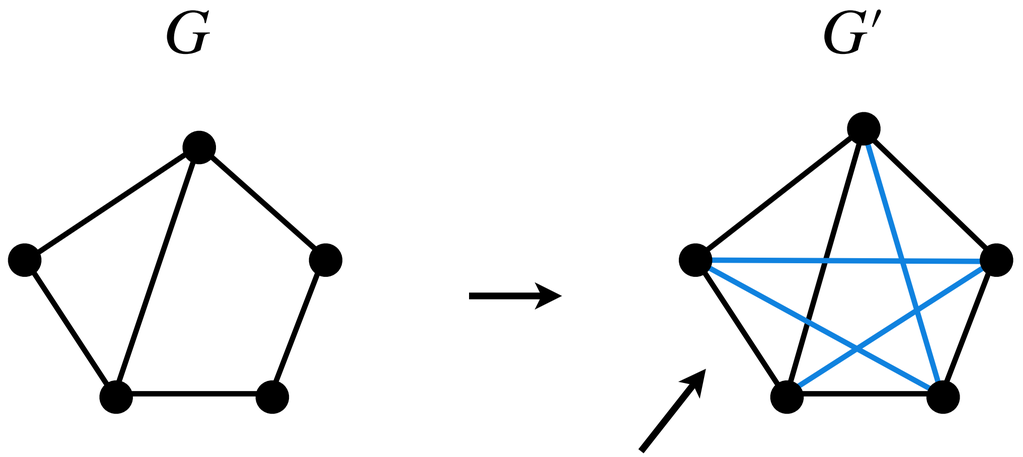

Blue edges have cost 1,
black edges have cost 0

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Claim: G contains a Hamiltonian cycle if and only if G' has TSP tour with cost ≤ 0.

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Claim: G contains a Hamiltonian cycle if and only if G' has TSP tour with cost ≤ 0.

Proof: ($\Longrightarrow$) If G contains a hamiltonian cycle C, then C in G' will be a TSP tour with cost 0.

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Then, every edge of C will have cost 0.

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Therefore, C is also a hamiltonian cycle in G.

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We know *VertexCover* is NP-complete.

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We can add $k - |S|$ more vertices to S to get a vertex cover of size k.

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While polytime exact algorithm is unlikely,
we may still have a decent polynomial approximation algorithm.

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Approximation Algorithms

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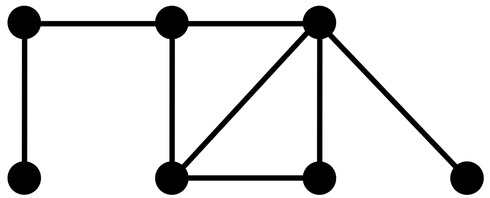

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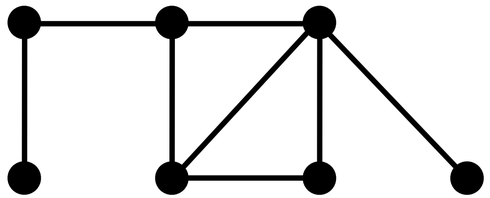

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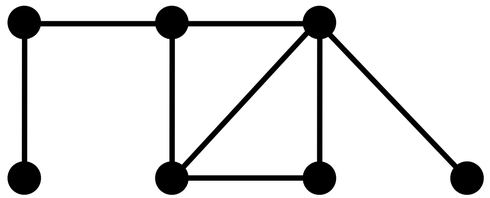

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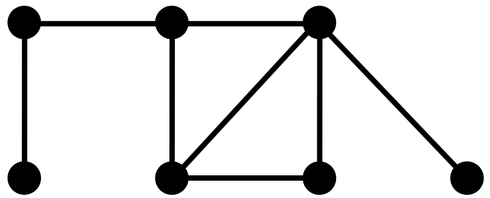

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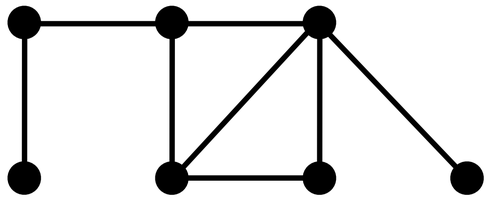

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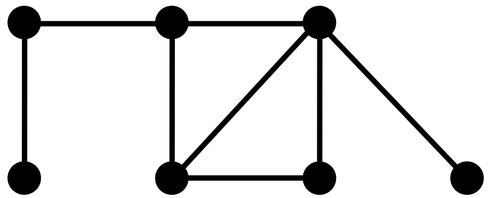

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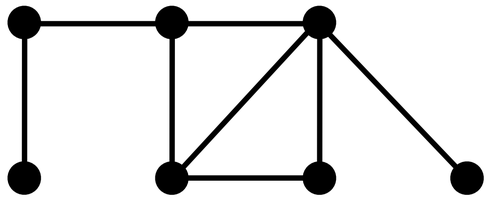

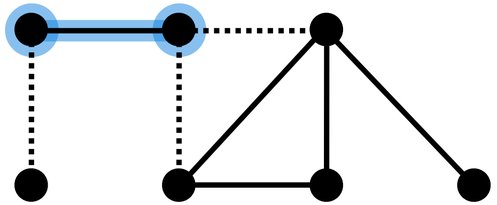

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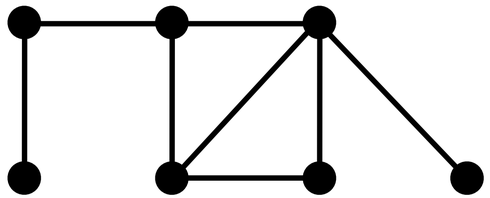

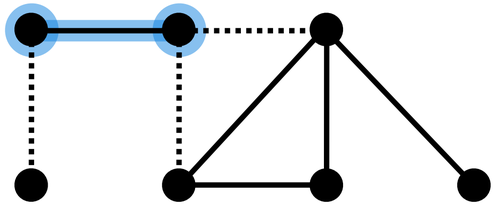

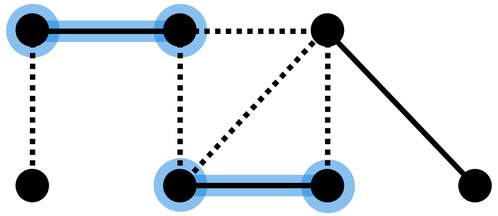

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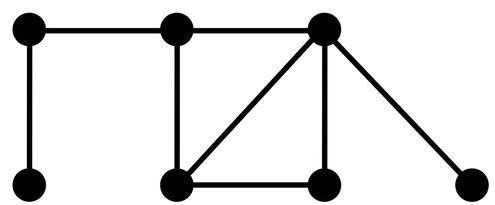

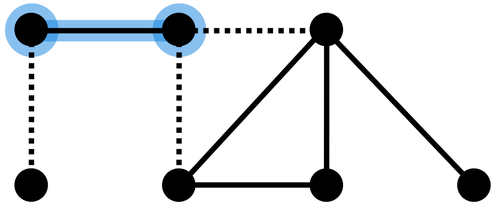

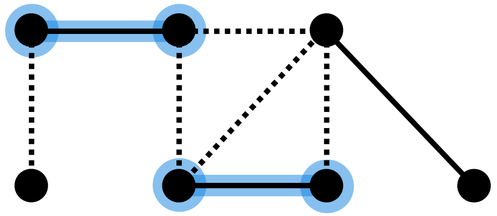

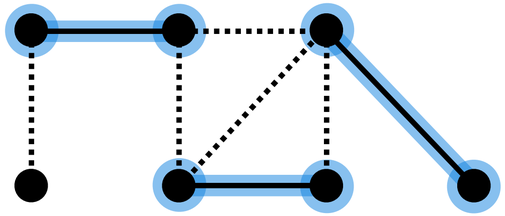

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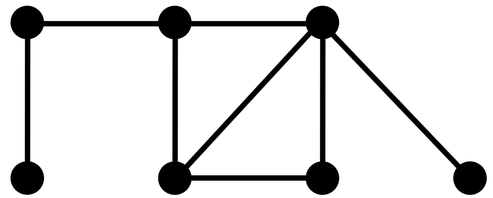

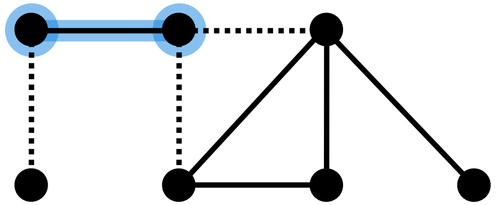

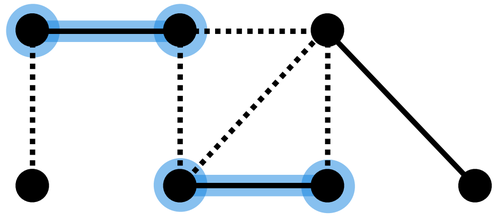

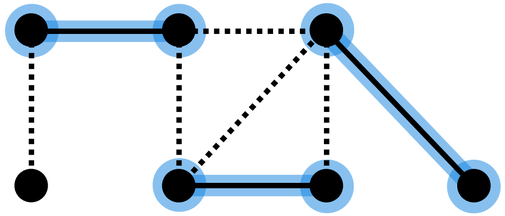

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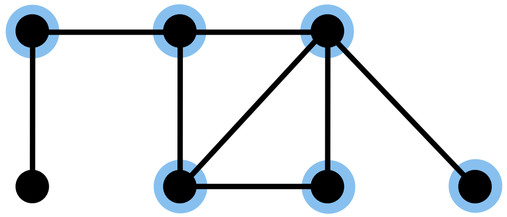

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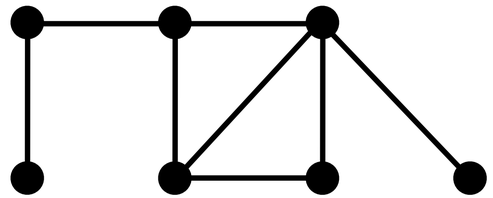

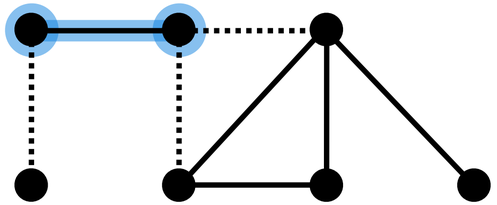

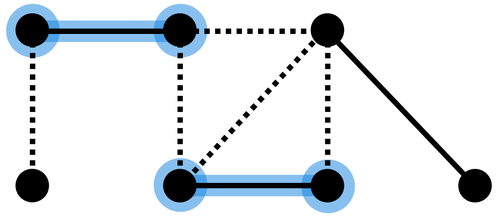

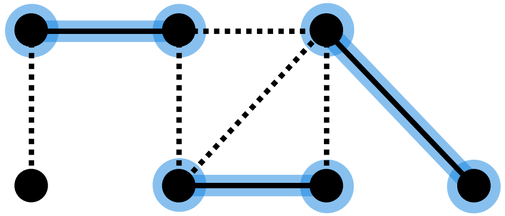

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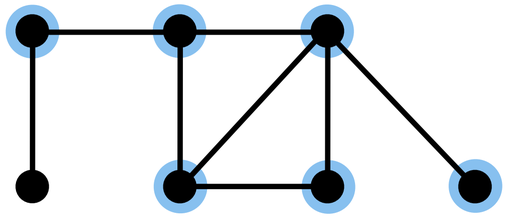

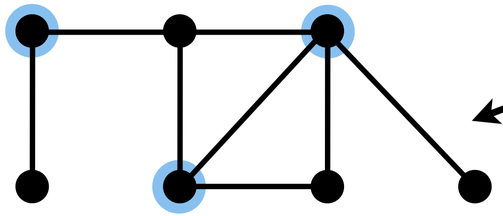

Minimum size vertex cover

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Approx-VC(G)

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Time Complexity: Polynomial in the size of G.

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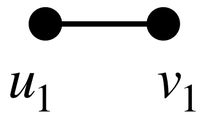

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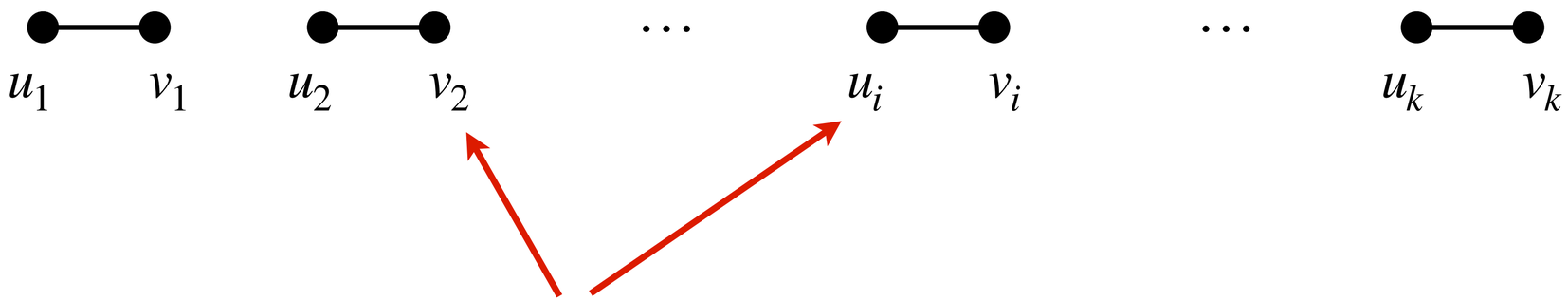

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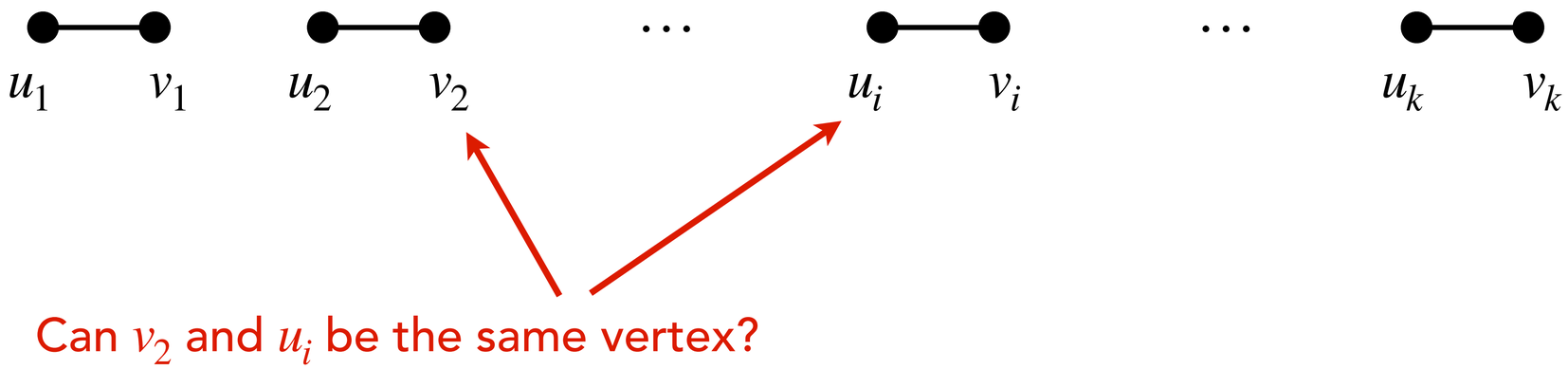

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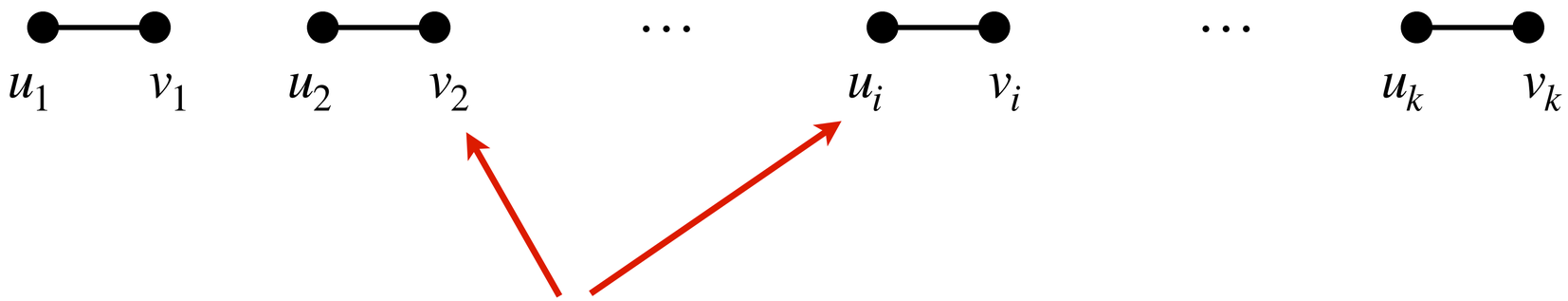

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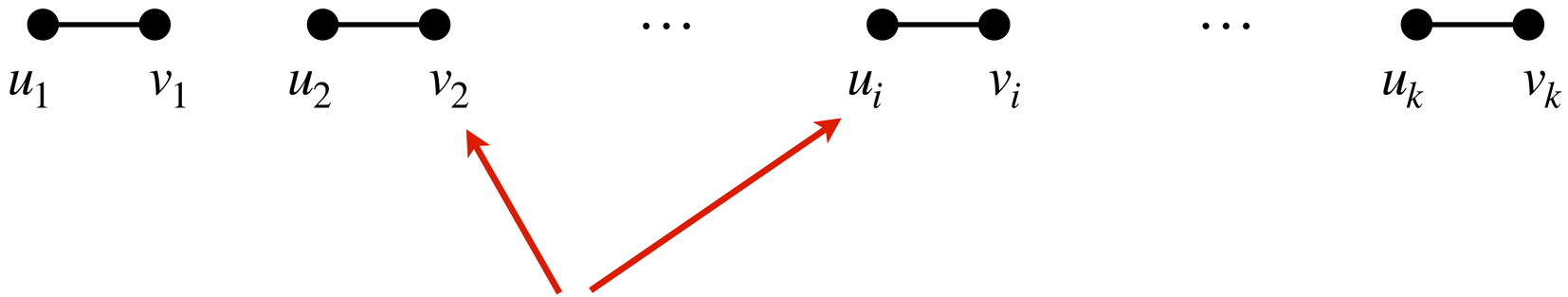

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